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论 文 量子密码和量子计算专题

基于角度的量子异常检测算法

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摘要 异常检测是机器学习和数据挖掘的一项重要任务, 在金融欺诈检测、网络入侵检测和医疗保健等领域发挥着关键作用. 基于角度的异常检测(Angle-Based Outlier Detection, ABOD)算法是目前最常用的异常检测方法之一, 其核心步骤是计算每个样本点与其他点的距离向量之间的角度方差, 并将角度方差低于某个阈值的点标记为异常点. 经典ABOD算法的时间复杂度关于样本数量M的规模为$O(M^3)$, 这使得该算法难以高效处理大规模数据集. 为此, 本文提出一个量子ABOD算法, 该算法利用量子幅度估计技术, 以量子并行方式计算所有点的角度方差, 并进一步利用量子幅度放大技术搜索出异常点. 所提量子ABOD算法的时间复杂度关于M的规模为$O\left[\sqrt{M}\log(M)\right]$, 与经典ABOD算法相比实现了$M$的多项式级加速.

关键词 基于角度的异常检测, 量子算法, 多项式级加速

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1 引言

量子计算利用量子叠加、量子纠缠等量子力学原理来完成计算任务. 与经典计算相比, 量子计算在解决某些特定问题上具有理论上的计算优势, 如大整数分解[1]、非结构搜索[2–4]以及求解线性方程组[5]. 近年来, 研究者们逐渐将其引入到机器学习领域, 从而诞生了一个新的学科研究领域——量子机器学习[6]. 量子机器学习自诞生以来, 已成为一个蓬勃发展的研究领域, 受到世界各国的广泛关注, 在数据分类[7–14]、聚类[15–17]、线性回归[18–23]、联邦学习[24–27]、关联规则挖掘[28,29]、降维[30–35]等多种机器学习任务上已取得了丰硕成果.

异常检测也是一项重要的机器学习任务, 它致力于解决如何识别那些不符合预期行为模式的数据或事件实例的问题[36], 被广泛应用于多个领域, 如金融欺诈检测[37]、网络入侵检测[38]和医疗保健[39]. 基于角度的异常检测(Angle-Based Outlier Detection, ABOD)算法[40]是最常用的异常检测方法之一, 其核心在于通过计算每个点与其他点的距离向量之间的角度方差, 即基于角度的异常因子(Angle-Based Outlier Factor, ABOF)来评估每个样本点的异常度, ABOF低

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于某个人为设定的阈值δ的样本点将标记为异常点. 在处理高维数据时, 相较于传统的基于距离度量的异常检测算法, ABOD算法展现出了更为优越的性能表现. 然而, 在处理大规模数据集时, ABOD算法的计算复杂度构成一个显著的挑战. 具体而言, 该算法计算每个样本的ABOF得分时, 其时间复杂度与样本数量的平方成正比, 而考虑到整个数据集上的所有样本, 整体的时间复杂度则上升为样本数量的三次方. 这一高昂的计算成本限制了ABOD算法在处理大规模数据集时的效率. 因此, 利用量子计算来降低ABOD的计算复杂度具有重要意义.

人们已经提出了一系列用于异常检测的量子算法, 并获得不同程度的量子优势[41]. 2018年, Liu和Rebentrost[42]提出量子异常检测算法的概念, 并设计了一个量子核主成分分析算法和一个量子单类支持向量机算法, 与对应的经典算法相比, 这两个算法在训练数据点数量和维度上均有指数级加速. 2019年, Liang等人[43]提出了一个基于密度估计的量子异常检测算法, 并指出该算法具有潜在的指数加速. 后来, Guo等人[44]改进了该算法, 并提出了一个新的基于密度估计的量子算法, 与经典算法相比, 该算法在数据点数量上实现了指数级加速. 2023年, Guo等人[45]提出了一个序列异常检测的量子算法, 在序列数量上实现了多项式级加速. 同年, Guo等人[46]又提出了一种基于局部异常因子(Local Outlier Factor, LOF)的量子异常检测算法, 相较于经典算法, 其在数据点数量上实现了多项式级加速, 在数据点维度上实现了指数加速. 尽管针对异常检测的量子算法已经取得如上所述的显著进展, 该领域的发展仍然处于初始阶段, 包括ABOD算法在内的许多异常检测问题仍无快速解决的量子算法, 该方向有待于进一步研究.

为了高效实现ABOD, 本文提出了一个量子ABOD算法. 参照于经典ABOD算法, 所提量子ABOD算法包括三个阶段. 首先, 由于ABOD的核心在于计算由三个不同样本点构成的夹角方差, 设计了一个量子子程序, 用于标记三个索引寄存器是否各自指代不同的样本点. 其次, 使用幅度估计[47]和量子算术(量子乘法器、量子加法器等)[48–50]并行计算距离向量的模以及两两间的内积. 最后, 在前两个阶段的基础上, 利用幅度估计并行获得每个样本点的ABOF得分, 并执行量子幅度放大算法[47]找寻所有的异常点. 与经典算法相比, 所提量子ABOD算法在样本点数量上可以实现多项式级加速.

2 经典ABOD算法回顾

本节简要回顾经典ABOD算法主要思想和流程, 更详细的介绍可见文献[40].

给定一个包含M个无标签样本点数据集$\mathcal{D} = \{x^i\}_{i=0}^{M-1}$, 其中$x^i = (x_1^i, x_2^i, \cdots, x_d^i) \in R^d$. 经典ABOD算法通过比较数据点与其他点之间距离向量的角度, 而非直接的距离, 来识别异常点. 考虑图1所示的一个简单数据集, 大部分非异常点多是在某一方向上的聚类簇内, 簇内的点四周各个方向都有数据点, 所以它和其他点之间距离向量角度变化较大, 即角度方差较大; 而聚类簇内的点相对于异常点都在某一个具体的方向上, 因此异常点的角度方差会相对较小. 基于此思想, ABOD算法采用距离加权的角度方差作为离群点值, 即ABOF值. 通过计算$\mathcal{D}$中所有样本点x^i的ABOF值, 并将其与预先设定的阈值δ进行比较, 以确定$\mathcal{D}$中所有的异常点. 整个过程描述如下.

(1) 计算数据点x^i与其他点之间的距离向量$x^{ij} = x^j - x^i$, $x^j \in \mathcal{D} \setminus \{x^i\}$.

(2) 计算距离向量的模$\|x^{ij}\|$和两两之间的内积$\langle x^{ij}, x^{ik}\rangle$:

$$\|x^{ij}\| = \sqrt{\sum_{l=0}^{d-1}\left(x_l^{ij}\right)^2}, \quad \langle x^{ij}, x^{ik}\rangle = \sum_{l=0}^{d-1} x_l^{ij} x_l^{ik}. \tag{1}$$

(3) 计算数据点x^i的ABOF值:

$$\begin{aligned}\mathrm{ABOF}(i) =& VAR_{x^j, x^k \in \mathcal{D}}\left(\frac{\langle x^{ij}, x^{ik}\rangle}{\|x^{ij}\|^2 \cdot \|x^{ik}\|^2}\right) \\ =& \frac{\sum_{x^j\in\mathcal{D}}\sum_{x^k\in\mathcal{D}} \frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|} \cdot \left(\frac{\langle x^{ij}, x^{ik}\rangle}{\|x^{ij}\|^2\cdot\|x^{ik}\|^2}\right)^2}{\sum_{x^j\in\mathcal{D}}\sum_{x^k\in\mathcal{D}} \frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|}} \\ &- \left(\frac{\sum_{x^j\in\mathcal{D}}\sum_{x^k\in\mathcal{D}} \frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|} \cdot \frac{\langle x^{ij}, x^{ik}\rangle}{\|x^{ij}\|^2\cdot\|x^{ik}\|^2}}{\sum_{x^j\in\mathcal{D}}\sum_{x^k\in\mathcal{D}} \frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|}}\right)^2,\end{aligned} \tag{2}$$

注意, 式中x^i, x^j, x^k为不同的点. 具体而言, $x^j \in \mathcal{D} \setminus \{x^i\}$并且$x^k \in \mathcal{D} \setminus \{x^i, x^j\}$.

(4) ABOF值同时考虑了数据点与其他点构成的角度范围以及距离. 一个较低的ABOF值意味着该数据

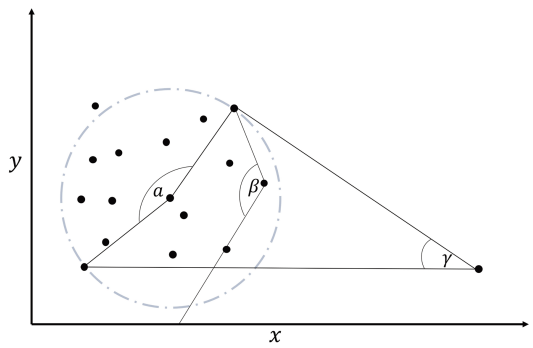

图 1 面向二维数据集的ABOD算法的一个简单例子

Figure 1 A simple example of ABOD algorithm for a two-dimensional dataset.

点与其他点构成的角度范围较小, 且距离较大, 这通常表明该数据点更可能是异常点. 因此, 提前设置一个阈值δ. 如果ABOF$(i) < \delta$, 标记数据点x^i为异常点; 否则标记为正常的数据点.

计算一个数据点的ABOF值的复杂度为$O(M^2d)$, 考虑所有M个点, 经典ABOD算法整体时间复杂度为$O(M^3d)$. 这意味着ABOD算法在处理大规模数据集时难以高效执行. 为此, 接下来提出一个高效的量子ABOD 算法.

3 量子ABOD算法

本节给出量子ABOD算法的具体过程, 并详细分析其复杂度.

假设数据集$\mathcal{D} = \{x^i\}_{i=0}^{M-1}$已经被存储在量子随机存储器(Quantum Random Access Memory, QRAM)[51]中, 使得能在$O[\log(Md)]$时间内有效地执行酉算子:

$$O_{\mathcal{D}} : |i\rangle |l\rangle |0\rangle \to |i\rangle |l\rangle \left|x_l^i\right\rangle, \tag{3}$$

其中, $i = 0, 1, \cdots, M-1$, $l = 0, 1, \cdots, d-1$.

量子算法由5个步骤组成. 步骤1中, 构造酉算子$U_{3-s} : |i\rangle |j\rangle |k\rangle |0\rangle \to |i\rangle |j\rangle |k\rangle \left|\mathrm{flag}(i, j, k)\right\rangle$, 其中$\left|\mathrm{flag}(i, j, k)\right\rangle$用来标记$x^i, x^j, x^k$是否为不同的点. 换句话说, 对任意$i = 0, 1, \cdots, M-1$, 定义集合:

$$\mathcal{S}_i := \{(j, k) | j, k \in \{0, 1, \cdots, M-1\}, j \neq i, k \neq i, j \neq k\}.$$

当$(j, k) \in \mathcal{S}_i$时, $\mathrm{flag}(i, j, k) = 1$, 否则$\mathrm{flag}(i, j, k) = 0$. 在步骤2和3中, 以量子并行方式计算模$\left\|\left\|x^{ij}\right\|\right\rangle$以及两两间的内积$\left|\left\langle x^{ij}, x^{ik}\right\rangle\right\rangle$; 在步骤4中, 通过量子算术以及幅度估计制备量子态$\frac{1}{\sqrt{M}} \sum_{i=0}^{M-1} |i\rangle |\mathrm{ABOF}(i)\rangle$. 在步骤5中, 执行量子幅度放大算法找到所有满足ABOF$(i) < \delta$的异常点.

3.1 算法

步骤1 实现酉操作

$$U_{3-s} : |i\rangle |j\rangle |k\rangle |0\rangle \to |i\rangle |j\rangle |k\rangle \left|\mathrm{flag}(i, j, k)\right\rangle,$$

其中, $i, j, k \in \{0, 1, \cdots, M-1\}$.

(1) 制备8个量子寄存器, 并初始化量子态为

$$|i\rangle |j\rangle |k\rangle |0\rangle^{\otimes m} |0\rangle |0\rangle |0\rangle |0\rangle, \tag{4}$$

其中前3个寄存器为索引寄存器, 第4寄存器为辅助寄存器, 后4个寄存器为单量子比特标记寄存器, $m = \lceil \log M \rceil$.

(2) 对第1和第4寄存器以及第2和第4寄存器各比特位分别执行CNOT门得到量子态:

$$|i\rangle |j\rangle |k\rangle \bigotimes_{l=0}^{m} |i_l \oplus j_l\rangle |0\rangle |0\rangle |0\rangle |0\rangle, \tag{5}$$

其中, $\oplus$表示异或符号, i_l和j_l分别表示i和j二进制形式的第l位.

(3) 对第4和第5寄存器执行多比特受控非门, 并执行逆操作还原第4寄存器得到量子态:

$$|i\rangle |j\rangle |k\rangle |0\rangle^{\otimes m} \left|\delta_{ij}\right\rangle |0\rangle |0\rangle |0\rangle, \tag{6}$$

其中, δ_{ij}表示克罗内克(Kronecker)符号, 即如果$i \neq j$, $\delta_{ij} = 0$, 否则$\delta_{ij} = 1$. 基于(1)–(3), 能构造酉算子$U_s : |i\rangle |j\rangle |0\rangle \to |i\rangle |j\rangle \left|\delta_{ij}\right\rangle$, 其量子线路如图 2所示.

(4) 分别对寄存器1, 3, 6以及寄存器2, 3, 7执行U_s得到量子态:

$$|i\rangle |j\rangle |k\rangle |0\rangle^{\otimes m} \left|\delta_{ij}\right\rangle |\delta_{ik}\rangle \left|\delta_{jk}\right\rangle |0\rangle. \tag{7}$$

(5) 对4个标记寄存器执行多比特受控非门, 并执行逆操作还原前3个标记寄存器, 得到量子态:

$$|i\rangle |j\rangle |k\rangle \left|\mathrm{flag}(i, j, k)\right\rangle. \tag{8}$$

至此, 实现了酉操作U_{3-s}, 其完整量子线路如图 3所示.

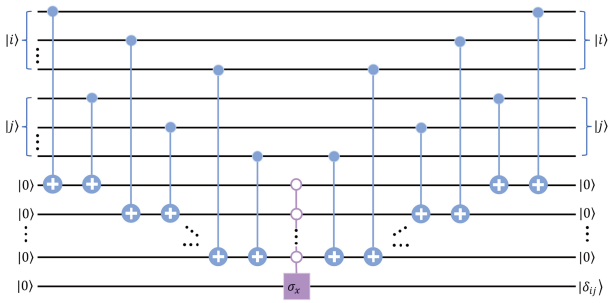

图 2 (网络版彩图) U_s量子线路图, 其中σ_x表示泡利**X**矩阵

Figure 2 (Color online) Quantum circuit of U_s, where σ_x represents the Pauli **X** matrix.

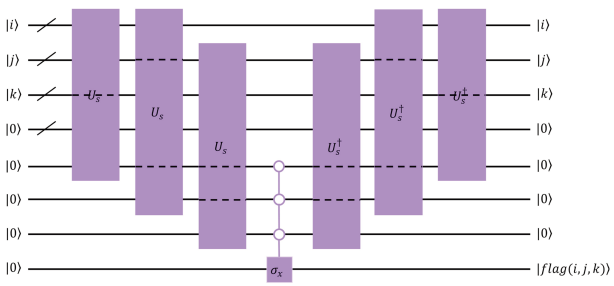

图 3 (网络版彩图) U_{3-s}量子线路图, 其中虚线表示酉算子未作用在该量子寄存器上

Figure 3 (Color online) Quantum circuit of U_{3-s}, where dashed lines indicate that the unitary operator does not act on the corresponding quantum register.

步骤2 实现酉操作

$$U_{ip}: |i\rangle|j\rangle|k\rangle|0\rangle \to |i\rangle|j\rangle|k\rangle\left|\left\langle x^{ij}, x^{ik}\right\rangle\right\rangle.$$

(1) 制备8个量子寄存器, 并初始化量子态为

$$|i\rangle_1|j\rangle_2|k\rangle_3\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)_4\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle_5|0\rangle_6|0\rangle_7|0\rangle_8. \tag{9}$$

(2) 执行黑盒$O_{\mathcal{D}}$, 得到量子态:

$$|i\rangle_1|j\rangle_2|k\rangle_3\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)_4\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle_5\left|x_l^i\right\rangle_6\left|x_l^j\right\rangle_7\left|x_l^k\right\rangle_8. \tag{10}$$

(3) 增加一个寄存器(寄存器9), 执行量子加法器[48]和乘法器[49,50]并行计算$x_l^{ij}x_l^{ik}=(x_l^j-x_l^i)(x_l^k-x_l^i)$, 得到量子态:

$$|i\rangle_1|j\rangle_2|k\rangle_3\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)_4\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle_5\left|x_l^{ij}x_l^{ik}\right\rangle_9. \tag{11}$$

(4) 增加一个辅助量子比特, 记作量子寄存器a, 执行$|0\rangle_4\langle 0|_4\otimes U_{9a}+|1\rangle_4\langle 1|_4\otimes I_{9a}$, 其中$U_{9a}$为受控旋转操作[8,52], 其基于$\left|x_l^{ij}x_l^{ik}\right\rangle_9$将$|0\rangle$旋转至$x_l^{ij}x_l^{ik}/C^2|0\rangle+\sqrt{1-(x_l^{ij}x_l^{ik}/C^2)^2}|1\rangle$, 这里$C=\max\{x_l^{ij}\}$. 接着, 执行(2)和(3)逆操作消除$x_l^i$, x_l^j, x_l^k和$x_l^{ij}x_l^{ik}$, 式(11)量子态将演化为

$$\begin{aligned}&|i\rangle_1|j\rangle_2|k\rangle_3\frac{1}{\sqrt{2}}\left[|0\rangle\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle\left(\frac{x_l^{ij}x_l^{ik}}{C^2}|0\rangle\right.\right.\\&\left.\left.+\sqrt{1-\left(\frac{x_l^{ij}x_l^{ik}}{C^2}\right)^2}|1\rangle\right)+|1\rangle\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle|0\rangle\right]_{45a}\\&:=|i\rangle_1|j\rangle_2|k\rangle_3\frac{1}{\sqrt{2}}\left(|0\rangle\left|\psi_{ijk}\right\rangle+|1\rangle\left|h_{ijk}\right\rangle\right)_{45a},\end{aligned} \tag{12}$$

其中, $\left|\psi_{ijk}\right\rangle=\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle\left(\frac{x_l^{ij}x_l^{ik}}{C^2}|0\rangle+\sqrt{1-\left(\frac{x_l^{ij}x_l^{ik}}{C^2}\right)^2}|1\rangle\right)$, $\left|h_{ijk}\right\rangle=\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle|0\rangle$.

(5) 对寄存器4执行Hadmard门, 得到量子态:

$$\begin{aligned}&|i\rangle_1|j\rangle_2|k\rangle_3\frac{1}{2}\left[|0\rangle\left(\left|\psi_{ijk}\right\rangle+\left|h_{ijk}\right\rangle\right)+|1\rangle\left(\left|\psi_{ijk}\right\rangle-\left|h_{ijk}\right\rangle\right)\right]_{45a}\\&:=|i\rangle_1|j\rangle_2|k\rangle_3\left|\phi_{ijk}\right\rangle_{45a},\end{aligned} \tag{13}$$

这里, 每个量子态$\left|\phi_{ijk}\right\rangle$可以写成$\left|\phi_{ijk}\right\rangle=\sin\theta_{ijk}\left|\phi_{ijk}^0\right\rangle+\cos\theta_{ijk}\left|\phi_{ijk}^1\right\rangle$, 其中$\left|\phi_{ijk}^0\right\rangle$和$\left|\phi_{ijk}^1\right\rangle$分别表示$|0\rangle\left(\left|\psi_{ijk}\right\rangle+\left|h_{ijk}\right\rangle\right)/2$以及$|1\rangle\left(\left|\psi_{ijk}\right\rangle-\left|h_{ijk}\right\rangle\right)/2$归一化后的量子态. 另外, $\sin^2\theta_{ijk}=\frac{1}{2}\left[1+\frac{1}{dC^2}\sum_{l=0}^{d-1}(x_l^{ij}x_l^{ik})\right]$, $\theta_{ijk}\in\left[0,\frac{\pi}{2}\right]$, 因此能通过幅度估计操作估计出$\sum_{l=0}^{d-1}x_l^{ij}x_l^{ik}$. 定义Grover算子$Q_{ijk}=-A_{ijk}S_0A_{ijk}^\dagger S_f$, 其中$A_{ijk}:|0\rangle_{45a}\to\left|\phi_{ijk}\right\rangle$, $S_0=I-2|0\rangle_{45a}\langle 0|_{45a}$, $S_f=(I-2|0\rangle_4\langle 0|_4)\otimes I$. Q_{ijk}有两个特征值$\lambda_\pm=\mathrm{e}^{\pm 2\mathrm{i}\theta_{ijk}}$和两个对应的特征向量$\left|\phi_{ijk}^\pm\right\rangle$.

(6) 增加辅助量子比特, 记为寄存器b, 利用算子Q_{ijk}执行幅度估计得到量子态:

$$|i\rangle_1|j\rangle_2|k\rangle_3\left(\left|\phi_{ijk}^+\right\rangle\left|\frac{\theta_{ijk}}{\pi}\right\rangle+\left|\phi_{ijk}^-\right\rangle\left|1-\frac{\theta_{ijk}}{\pi}\right\rangle\right)_{45ab}. \tag{14}$$

(7) 通过量子乘法器以及sine门[50]计算$\left|\left\langle x^{ij},x^{ik}\right\rangle\right\rangle=\left|dC^2(2\sin^2\theta_{ijk}-1)\right\rangle$, 执行逆操作还原其余寄存器得到量子态:

$$|i\rangle_1|j\rangle_2|k\rangle_3\left|\left\langle x^{ij},x^{ik}\right\rangle\right\rangle_b. \tag{15}$$

步骤3 实现酉操作

$$U_{\text{norm}}: |i\rangle|j\rangle|k\rangle|0\rangle \to |i\rangle|j\rangle|k\rangle\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle.$$

(1) 根据步骤2中(2)和(3), 制备初态:

$$|i\rangle|j\rangle\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle\left|x_l^{ij}\right\rangle. \tag{16}$$

(2) 增加一个辅助量子比特执行受控旋转, 得到量子态:

$$|i\rangle|j\rangle\frac{1}{\sqrt{d}}\sum_{l=0}^{d-1}|l\rangle\left(\frac{x_l^{ij}}{C}|0\rangle+\sqrt{1-\left(\frac{x_l^{ij}}{C}\right)^2}|1\rangle\right), \tag{17}$$

其中最后辅助量子比特的量子态中$|0\rangle$的幅度平方和为$\sum_{l=0}^{d-1}\left(\frac{x_l^{ij}}{C\sqrt{d}}\right)^2=\frac{\left\|x^{ij}\right\|^2}{dC^2}:=\sin^2(\theta_{ij})$.

(3) 类似于步骤2中(6)和(7), 执行幅度估计, 再通过量子乘法器和sine门计算$\left|\left\|x^{ij}\right\|\right\rangle=\left|\sqrt{d}C\sin\theta_{ij}\right\rangle$, 得到量子态:

$$|i\rangle|j\rangle\left|\left\|x^{ij}\right\|\right\rangle. \tag{18}$$

(4) 针对i和k重复(1)–(3), 得到量子态:

$$|i\rangle|j\rangle|k\rangle\left|\left\|x^{ij}\right\|\right\rangle\left|\left\|x^{ik}\right\|\right\rangle. \tag{19}$$

(5) 通过量子乘法器, 得到量子态:

$$|i\rangle|j\rangle|k\rangle\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle. \tag{20}$$

步骤4 实现酉操作

$$U_{\text{abof}}: |i\rangle|0\rangle \mapsto |i\rangle|\text{ABOF}(i)\rangle,$$

对任意的$i\in\{0,1,\cdots,M-1\}$.

(1) 制备量子初态:

$$|i\rangle\frac{1}{M}\sum_{j,k=0}^{M-1}|j\rangle|k\rangle|0\rangle|0\rangle. \tag{21}$$

(2) 执行U_{3-s}和U_{norm}, 得到量子态:

$$|i\rangle\frac{1}{M}\sum_{j,k=0}^{M-1}|j\rangle|k\rangle\left|\text{flag}(i,j,k)\right\rangle\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle. \tag{22}$$

(3) 增加一个量子比特执行受控旋转并执行U_{norm}的逆操作消除$\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle$, 得到量子态:

$$\begin{aligned}&|i\rangle\frac{1}{M}\sum_{j,k=0}^{M-1}|j\rangle|k\rangle\left|\text{flag}(i,j,k)\right\rangle\left(\frac{E}{\sqrt{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}}|0\rangle\right.\\&\left.+\sqrt{1-\frac{E^2}{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}}|1\rangle\right)\\&:=|i\rangle\frac{1}{M}\sum_{(j,k)\in\mathcal{S}_i}|j\rangle|k\rangle|1\rangle\left(\frac{E}{\sqrt{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}}|0\rangle\right.\\&\left.+\sqrt{1-\frac{E^2}{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}}|1\rangle\right)+|\omega_1\rangle,\end{aligned} \tag{23}$$

其中, $E=\min_{i\neq j}\left\{\left\|x^{ij}\right\|\right\}$, $|\omega_1\rangle$以及之后符号包含ω的量子态表示为$\text{flag}(i,j,k)=0$部分未归一化的量子叠加态. 注意, 这里受控旋转只需受控于$\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle$, 无需受控于$\left|\text{flag}(i,j,k)\right\rangle$, 因为这不会对后续操作产生影响.

(4) 由于式(23)叠加项中最后两个量子比特的量子态$|1\rangle|0\rangle$的幅度平方和为$\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{E^2}{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}:=\sin^2(\alpha_i)$, 执行幅度估计并行估计$\alpha_i$, 再通过量子乘法器和sine门计算$|p_i\rangle=\left|\frac{\sin^2\alpha_i}{E^2}\right\rangle$, 最后执行(1)–(3)的逆操作, 去除不必要的寄存器, 得到量子态:

$$|i\rangle\left|\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{1}{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}\right\rangle:=|i\rangle|p_i\rangle. \tag{24}$$

(5) 根据步骤1–3, 能制备量子初态:

$$|i\rangle\frac{1}{M}\sum_{j,k=0}^{M-1}|j\rangle|k\rangle\left|\text{flag}(i,j,k)\right\rangle\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle\left|\left\langle x^{ij},x^{ik}\right\rangle\right\rangle. \tag{25}$$

(6) 增加一个量子比特执行受控旋转, 得到量子态:

$$\begin{aligned}&|i\rangle\frac{1}{M}\sum_{(j,k)\in\mathcal{S}_i}|j\rangle|k\rangle|1\rangle\left(E^3\sqrt{\frac{\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}}|0\rangle\right.\\&\left.+\sqrt{1-\frac{E^6\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}}|1\rangle\right)+|\omega_2\rangle.\end{aligned} \tag{26}$$

(7) 类似(4), 式(26)叠加项中最后两个量子比特的量子态$|1\rangle|0\rangle$的幅度平方和为$\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{E^6\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}:=\sin^2(\beta_i)$, 执行幅度估计并行估计$\beta_i$, 再通过量子乘法器

和sine门计算$|q_i\rangle = \left|\frac{\sin^2\beta_i}{E^6}\right\rangle$, 执行(5)和(6)的逆操作并剔除不必要的量子寄存器, 得到量子态:

$$|i\rangle\left|\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}^{M}\frac{\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}\right\rangle := |i\rangle|q_i\rangle. \tag{27}$$

(8) 在(5)制备的量子态基础上, 增加一个初始态为$\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$的辅助量子比特, 记为寄存器$c$, 得到量子态:

$$|i\rangle\frac{1}{\sqrt{2}}(|0\rangle_c+|1\rangle_c)\frac{1}{M}\sum_{j,k=0}^{M-1}|j\rangle|k\rangle\left|\mathrm{flag}(i,j,k)\right\rangle$$
$$\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle\left|\left\langle x^{ij},x^{ik}\right\rangle\right\rangle. \tag{28}$$

(9) 在式(28)中$|k\rangle$和$\left|\mathrm{flag}(i,j,k)\right\rangle$之间再增加一个辅助量子比特, 并执行受控旋转操作, 再执行(8)逆操作去除寄存器$\left|\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|\right\rangle$和$\left|\left\langle x^{ij},x^{ik}\right\rangle\right\rangle$, 得到量子态:

$$\begin{aligned}&|i\rangle\frac{1}{\sqrt{2}}\left[|0\rangle_c\frac{1}{M}\sum_{(j,k)\in\mathcal{S}_i}^{M}|j\rangle|k\rangle\left(\frac{E^4\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^3\cdot\left\|x^{ik}\right\|^3}|0\rangle\right.\right.\\&\left.+\sqrt{1-\frac{E^8\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^6\cdot\left\|x^{ik}\right\|^6}}|1\rangle\right)\\&\left.+|1\rangle_c\frac{1}{M}\sum_{(j,k)\in\mathcal{S}_i}^{M}|j\rangle|k\rangle|0\rangle\right]|1\rangle_{\mathrm{flag}}+|\omega_3\rangle\\&:=|i\rangle\frac{1}{\sqrt{2}}\left[|0\rangle_c|\varphi_i\rangle+|1\rangle_c|\rho_i\rangle\right]|1\rangle_{\mathrm{flag}}+|\omega_3\rangle.\end{aligned} \tag{29}$$

为了后续方便识别, 式(29)中右下角的flag表示量子态$\left|\mathrm{flag}(i,j,k)\right\rangle$所在的标记寄存器.

(10) 对辅助量子比特c执行Hadmard门, 得到量子态:

$$|i\rangle\frac{1}{2}\left[|0\rangle_c(|\varphi_i\rangle+|\rho_i\rangle)+|1\rangle_c(|\varphi_i\rangle-|\rho_i\rangle)\right]|1\rangle_{\mathrm{flag}}+|\omega_4\rangle. \tag{30}$$

(11) 式(30)叠加项中量子比特c和标记寄存器的量子态$|0\rangle_c|1\rangle_{\mathrm{flag}}$的幅度平方和为$\frac{1}{2}D+\frac{E^4}{2M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\langle x^{ij},x^{ik}\rangle}{\|x^{ij}\|^3\cdot\|x^{ik}\|^3} := \sin^2(\gamma_i)$, 执行幅度估计并行估计$\gamma_i$, 并进一步并行计算$|o_i\rangle = \left|\frac{2\sin^2\gamma_i-D}{E^4}\right\rangle$, 其中$D=\frac{(M-1)(M-2)}{M^2}$, 得到量子态:

$$|i\rangle\left|\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^3\cdot\left\|x^{ik}\right\|^3}\right\rangle := |i\rangle|o_i\rangle. \tag{31}$$

(12) 结合(4), (7)和(11), 通过量子算术操作计算$|\mathrm{ABOF}(i)\rangle = \left|\frac{q_i}{p_i}-\frac{o_i^2}{p_i^2}\right\rangle$, 得到了如图1所示的任意样本点$x^i$ $(i=0,1,\cdots,M-1)$的ABOF值. 由于前面所有步骤均为酉操作, 因此实现了目标酉操作U_{abof}.

由于酉操作的线性, 取输入为$\frac{1}{\sqrt{M}}\sum_{i=0}^{M-1}|i\rangle$而非$|i\rangle$, 执行$U_{\mathrm{abof}}$得到所有样本点的ABOF值:

$$\frac{1}{\sqrt{M}}\sum_{i=0}^{M-1}|i\rangle|\mathrm{ABOF}(i)\rangle. \tag{32}$$

步骤5 执行量子幅度放大并测量, 搜索出所有满足$\mathrm{ABOF}(i)<\delta$的异常数据点.

整个算法量子线路如图4所示.

3.2 复杂度分析

(1) 步骤1的时间复杂度为$O(\log M)$, 具体分析如下.

在步骤1的(2)中, 分别执行$O(\log M)$个CNOT门. 步骤1中(3)执行了一个多比特受控非门, 其复杂度为$O(\log M)$. 步骤1中(4)再重复执行步骤1中(1)–(3)两次, 步骤1中(5)执行CCCNOT门, 其复杂度为$O(1)$. 因此步骤1的整体复杂度为$O(\log M)$.

(2) 步骤2和3的时间复杂度分别为$O[\log(Md)/\varepsilon_1]$和$O[\log(Md)/\varepsilon_2]$, 主要来源于黑盒$O_{\mathcal{D}}$以及幅度估计, 具体细节如下.

步骤2中(1)和(2)分别执行Hadmard门和$O_{\mathcal{D}}$, 时间复杂度分别为$O(\log d)$ 和$O[\log(Md)]$. 步骤2中(3)执行量子加法器和量子乘法器, 复杂度为$O[\mathrm{poly}(n)]$[49], 其中n为每个数据点的特征值存储精度, 即x_l^i二进制表示的比特数. 由于n的大小通常与M和d无关, 这里假定$n=O(1)$, 则这些门的复杂度可忽略不计. 步骤2中(4)和(5)执行受控旋转和一个Hadmard门, 复杂度为$O[\mathrm{poly}(n)]$. 在步骤2的(6)中, 为确保$\Delta\theta_{ijk}\leqslant\varepsilon_1$, 需连续执行$O(1/\varepsilon_1)$次算子$Q_{ijk}$. 而算子$Q_{ijk}$的复杂度为$O[\log(Md)]$, 因此幅度估计过程的复杂度为$O[\log(Md)/\varepsilon_1]$. 类似于步骤2中(3)的分析, 步骤2中(7)执行量子乘法器和sine门, 复杂度为$O[\mathrm{poly}(n)]$). 步骤3中(1)执行Hadmard门和$O_{\mathcal{D}}$, 复杂

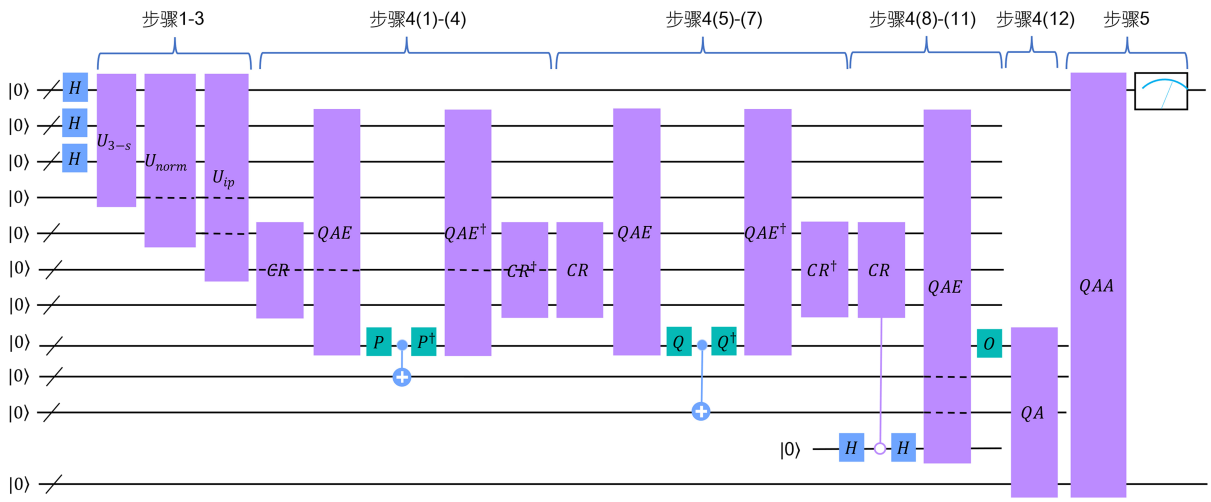

图 4 (网络版彩图)量子ABOD算法整体量子线路图. 其中U_{3-s}, U_{ip}和$U_{\rm norm}$分别表示步骤1–3的酉操作, CR表示受控旋转, QAE表示幅度估计, QAA表示幅度放大, P, Q, O和QA分别表示步骤4中(4), (7), (11)和(12)计算p_i, q_i, o_i和ABOF(i)的量子算术操作

Figure 4 (Color online) The overall quantum circuit diagram for the quantum ABOD algorithm. Here, U_{3-s}, U_{ip}, and $U_{\rm norm}$ represent the unitary operations for steps 1 to 3, respectively. CR denotes controlled rotations, QAE represents quantum amplitude estimation, and QAA represents quantum amplitude amplification. Furthermore, P, Q, O, and QA denote the quantum arithmetic operations for calculating p_i, q_i, o_i and ABOF(i) in (4), (7), (11), and (12) of step 4, respectively.

度为$O[\log(Md)]$. 步骤3中(2)执行受控旋转, 复杂度为$O[\mathrm{poly}(n)]$. 同理, 在步骤3的(3) 中, 为确保$\Delta\theta_{ij} \leqslant \varepsilon_2$, 幅度估计的复杂度为$O[\log(Md)/\varepsilon_2]$. 步骤3中(4)重复步骤3中(1)–(3), 其复杂度为$O[\log(Md)/\varepsilon_2]$. 步骤3中(5)执行量子乘法器, 复杂度为$O[\mathrm{poly}(n)]$.

接下来分析$\langle x^{ij}, x^{ik}\rangle$和$\|x^{ij}\|$的误差, 其主要来源于步骤2中(6)和步骤3中(3)的幅度估计. 为方便说明, 用$\hat{\alpha}$表示α的估计值, Δ表示误差, 即$\Delta a = |\hat{a} - a|$. 由于$\Delta\theta_{ijk} \leqslant \varepsilon_1$, 即$|\hat{\theta_{ijk}} - \theta_{ijk}| \leqslant \varepsilon_1$, 则

$$\begin{aligned}\left|\widehat{\langle x^{ij}, x^{ik}\rangle} - \langle x^{ij}, x^{ik}\rangle\right| &= 2dC^2\left|\sin^2\hat{\theta}_{ijk} - \sin^2\theta_{ijk}\right| \\ &\leqslant 2dC^2\varepsilon_1. \end{aligned} \tag{33}$$

同时, 由于$\Delta\theta_{ij} \leqslant \varepsilon_2$, 则

$$\left|\|\hat{x^{ij}}\| - \|x^{ij}\|\right| = \sqrt{d}C\left|\sin\hat{\theta}_{ij} - \sin\theta_{ij}\right| \leqslant \sqrt{d}C\varepsilon_2. \tag{34}$$

(3) 步骤4的复杂度主要来源于步骤4中(4), (7)和(11)的幅度估计部分. 假设这三步幅度估计求出的α_i, β_i 以及γ_i的误差分别为ε_3, ε_4以及ε_5, 则估计p_i, q_i 和o_i的复杂度分别为$O\left[\frac{\log(Md)}{\varepsilon_2\varepsilon_3}\right]$, $O\left[\frac{\log(Md)}{\varepsilon_1\varepsilon_4} + \frac{\log(Md)}{\varepsilon_2\varepsilon_4}\right]$和$O\left[\frac{\log(Md)}{\varepsilon_1\varepsilon_5} + \frac{\log(Md)}{\varepsilon_2\varepsilon_5}\right]$.

同时, p_i, q_i和o_i误差如下:

$$\begin{aligned}&\left|\hat{p}_i - \frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|}\right| \\ &\leqslant \left|\hat{p}_i - \frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{1}{\|\hat{x^{ij}}\|\cdot\|\hat{x^{ik}}\|}\right| \\ &+ \left|\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{1}{\|\hat{x^{ij}}\|\cdot\|\hat{x^{ik}}\|} - p_i\right| \\ &\leqslant \frac{\varepsilon_3}{E^2} + \frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\left|\frac{1}{\|\hat{x^{ij}}\|\cdot\|\hat{x^{ik}}\|} - \frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|}\right|, \end{aligned} \tag{35}$$

其中不等式(35)的第二项中求和部分:

$$\begin{aligned}&\left|\frac{1}{\|\hat{x^{ij}}\|\cdot\|\hat{x^{ik}}\|} - \frac{1}{\|x^{ij}\|\cdot\|x^{ik}\|}\right| \\ &\leqslant \left|\frac{\|x^{ik}\| - \|\hat{x^{ik}}\|}{\|\hat{x^{ij}}\|\cdot\|\hat{x^{ik}}\|\cdot\|x^{ik}\|}\right| + \left|\frac{\|x^{ij}\| - \|\hat{x^{ij}}\|}{\|\hat{x^{ij}}\|\cdot\|x^{ij}\|\cdot\|x^{ik}\|}\right| \\ &\leqslant \frac{2\sqrt{d}C\varepsilon_2}{E^3}. \end{aligned} \tag{36}$$

因此, p_i的误差为$\dfrac{\varepsilon_3}{E^2}+D\dfrac{2\sqrt{d}C\varepsilon_2}{E^3}$.

$$\begin{aligned}
&\left|\hat{q}_i-\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}\right|\\
&\leqslant\left|\hat{q}_i-\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle^2}{\left\|\hat{x^{ij}}\right\|^5\cdot\left\|\hat{x^{ik}}\right\|^5}\right|\\
&+\left|\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle^2}{\left\|\hat{x^{ij}}\right\|^5\cdot\left\|\hat{x^{ik}}\right\|^5}-q_i\right|\\
&\leqslant\frac{\varepsilon_4}{E^6}+\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\left|\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle^2}{\left\|\hat{x^{ij}}\right\|^5\cdot\left\|\hat{x^{ik}}\right\|^5}-\frac{\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}\right|,
\end{aligned}\tag{37}$$

其中不等式(37)的第二项中求和部分:

$$\begin{aligned}
&\left|\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle^2}{\left\|\hat{x^{ij}}\right\|^5\cdot\left\|\hat{x^{ik}}\right\|^5}-\frac{\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}\right|\\
&\leqslant\left|\frac{5\left\langle x^{ij},x^{ik}\right\rangle^2}{\left\|x^{ij}\right\|^4\cdot\left\|x^{ik}\right\|^4}\Delta\frac{1}{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}\right|\\
&+\left|\frac{2\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^5\cdot\left\|x^{ik}\right\|^5}\Delta\left\langle x^{ij},x^{ik}\right\rangle\right|\\
&\leqslant\frac{10\sqrt{d}C}{E^7}\varepsilon_2+\frac{4dC^2}{E^8}\varepsilon_1,
\end{aligned}\tag{38}$$

则q_i的误差为$\frac{\varepsilon_4}{E^6}+D\left(\frac{10\sqrt{d}C}{E^7}\varepsilon_2+\frac{4dC^2}{E^8}\varepsilon_1\right)$.

$$\begin{aligned}
&\left|\hat{o}_i-\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^3\cdot\left\|x^{ik}\right\|^3}\right|\\
&\leqslant\left|\hat{o}_i-\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle}{\left\|\hat{x^{ij}}\right\|^3\cdot\left\|\hat{x^{ik}}\right\|^3}\right|\\
&+\left|\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle}{\left\|\hat{x^{ij}}\right\|^3\cdot\left\|\hat{x^{ik}}\right\|^3}-o_i\right|\\
&\leqslant\frac{2\varepsilon_5}{E^4}+\frac{1}{M^2}\sum_{(j,k)\in\mathcal{S}_i}\left|\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle}{\left\|\hat{x^{ij}}\right\|^3\cdot\left\|\hat{x^{ik}}\right\|^3}-\frac{\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^3\cdot\left\|x^{ik}\right\|^3}\right|,
\end{aligned}\tag{39}$$

其中不等式(39)的第二项中求和部分:

$$\begin{aligned}
&\left|\frac{\left\langle\widehat{x^{ij},x^{ik}}\right\rangle}{\left\|\hat{x^{ij}}\right\|^3\cdot\left\|\hat{x^{ik}}\right\|^3}-\frac{\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^3\cdot\left\|x^{ik}\right\|^3}\right|\\
&\leqslant\left|\frac{3\left\langle x^{ij},x^{ik}\right\rangle}{\left\|x^{ij}\right\|^2\cdot\left\|x^{ik}\right\|^2}\Delta\frac{1}{\left\|x^{ij}\right\|\cdot\left\|x^{ik}\right\|}\right|\\
&+\left|\frac{1}{\left\|x^{ij}\right\|^3\cdot\left\|x^{ik}\right\|^3}\Delta\left\langle x^{ij},x^{ik}\right\rangle\right|\\
&\leqslant\frac{6\sqrt{d}C}{E^5}\varepsilon_2+\frac{2dC^2}{E^6}\varepsilon_1,
\end{aligned}\tag{40}$$

则o_i的误差为$\frac{2\varepsilon_5}{E^4}+D\left(\frac{6\sqrt{d}C}{E^5}\varepsilon_2+\frac{2dC^2}{E^6}\varepsilon_1\right)$.

最后, 对任一样本点x^i的ABOF值(ABOF(i))的误差进行如下分析:

$$\begin{aligned}
&\left|\widehat{\mathrm{ABOF}}(i)-\mathrm{ABOF}(i)\right|\\
&=\left|\left(\frac{\hat{q}_i}{\hat{p}_i}-\frac{\hat{o_i^2}}{\hat{p_i^2}}\right)-\left(\frac{q_i}{p_i}-\frac{o_i^2}{p_i^2}\right)\right|\\
&\leqslant\left|\left(\frac{2o_i^2}{p_i^3}-\frac{q_i}{p_i^2}\right)\Delta p_i\right|+\left|\frac{1}{p_i}\Delta q_i\right|+\left|\frac{2o_i}{p_i^2}\Delta o_i\right|\\
&\leqslant\frac{F^2}{E^6}\left(\frac{4\varepsilon_5+\varepsilon_4+\varepsilon_3}{D}+\frac{24\sqrt{d}C}{E}\varepsilon_2+\frac{8dC^2}{E^2}\varepsilon_1\right),
\end{aligned}\tag{41}$$

其中, $F=\max_{i\neq j}\{\left\|x^{ij}\right\|\}$, $\frac{1}{p_i}\leqslant\frac{F^2}{D}$, $\frac{q_i}{p_i}\leqslant\frac{1}{E^4}$, $\frac{o_i}{p_i}\leqslant\frac{1}{E^2}$. 如果$E,F,C=O(1)$, 并且$\varepsilon_1=\frac{E^8}{40dC^2F^2}\varepsilon$, $\varepsilon_2=\frac{E^7}{120\sqrt{d}CF^2}\varepsilon$, $\varepsilon_3=\varepsilon_4=\frac{DE^6}{5F^2}\varepsilon$, $\varepsilon_5=\frac{DE^6}{20F^2}\varepsilon$. 则ABOF($i$)的误差为$O(\varepsilon)$. 步骤4的复杂度为$O\left[d\log(Md)/\varepsilon^2\right]$.

(4) 步骤5的量子幅度放大和测量的复杂度为$O\left[d\sqrt{MT}\log(Md)/\varepsilon^2\right]$, 其中$T$ 为异常点数量. 一般而言, 异常点数量$T<<M$, 可忽略不计, 因此该算法的整体复杂度为$O\left[d\sqrt{M}\log(Md)/\varepsilon^2\right]$. 与对应的经典算法相比, 本文所提出的算法在样本数量M上实现了多项式加速.

4 结论

针对经典ABOD算法处理大规模数据集复杂度过高的问题, 本文提出了一个量子ABOD算法. 结果表明, 与经典算法相比, 所提量子算法在数据点数量M上可以实现多项式级加速. 当前存在多种多样的异常检测算法, 这些算法在经典计算领域已展现出强大性能, 它们有望扩展到量子领域, 但目前相关的研究依然太少. 本文所提出的量子ABOD算法有望启发更多的异常检测算法获得量子加速优势.

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Quantum algorithm for angle-based anomaly detection

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Anomaly detection is a prominent task in machine learning and data mining, which plays a key role in various domains such as financial fraud detection, network intrusion detection, and healthcare. Angle-based outlier detection (ABOD) algorithm stands out as one of the most common approaches for anomaly detection, the core step of which involves computing the angular variance among distance vectors from each point to others, and then marking those points with angular variance lower than a threshold as outliers. The time complexity of classical ABOD algorithm scales as $O(M^3)$ with respect to the number of samples M, making it hard to efficiently handle large-scale datasets. To this end, a quantum ABOD algorithm is proposed, in which quantum amplitude estimation is harnessed to compute angular variance in quantum parallel, and quantum amplitude amplification is further applied to search out the outliers. Compared with its classical counterpart, our quantum ABOD algorithm has time complexity $O\left[\sqrt{M}\log(M)\right]$ with respect to M, achieving a polynomial speedup with respect to M.

angle-based outlier detection, quantum algorithm, polynomial speedup

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